PART 3 - CAPSTONE PROJECT #6

Glossary & 15-Question Quiz

Task 66: Key Definitions, 15 MCQs, and Technical Answer Key.

1. Glossary

AM
Carrier amplitude varies with message signal voltage.

FM
Carrier frequency varies with message signal amplitude.

ASK
Digital keying using carrier ON/OFF states for binary 1/0.

FSK
Digital keying using two discrete frequencies for binary bits.

2. Assessment Quiz (Q1 to Q6)

Q1. In AM, which parameter of the carrier wave is varied?

A) Frequency
B) Phase
C) Amplitude
D) Wavelength

Q2. What is the required bandwidth for standard AM signal with modulating frequency f_m?

A) f_m
B) $2 \times f_m$
C) $4 \times f_m$
D) $f_m / 2$

Q3. Why is FM less susceptible to noise than AM?

A) Lower frequency
B) Noise affects amplitude, not frequency
C) Uses less power
D) Less bandwidth

Q4. What is Carson's Rule for FM Bandwidth?

A) $2(\Delta f + f_m)$
B) $\Delta f + 2f_m$
C) $2 \times \Delta f$
D) $\Delta f \times f_m$

Q5. Which scheme is also called On-Off Keying (OOK)?

A) FSK
B) Binary ASK
C) PSK
D) QAM

Q6. In FSK, signal amplitude during bit transmission is:

A) Doubled
B) Zero
C) Constant
D) Inverted

2. Assessment Quiz (Q7 to Q15)

Q7. Overmodulation in AM occurs when modulation index μ is:

A) < 0.5 B) > 1.0

C) Equal to 0 D) Equal to 0.5

Q8. If f_m = 5 kHz in AM, what is the bandwidth?

A) 5 kHz B) 10 kHz

C) 20 kHz D) 2.5 kHz

Q9. Which parameter remains constant in FM?

A) Instantaneous frequency B) Peak amplitude

C) Phase angle D) Bit duration

Q10. Standard bandwidth for ASK with bit rate R_b is:

A) R_b B) $R_b/2$

C) $2\ R_b$ D) $4\ R_b$

Q11. Modulation scheme used in Caller ID and legacy modems:

A) AM B) FSK

C) ASK D) QAM

Q12. FM index β if Δf = 75 kHz and f_m = 15 kHz is:

A) 0.2 B) 5

C) 90 D) 1125

Q13. Optical fiber pulses commonly use:

A) ASK / OOK B) Wideband FM

C) FSK D) AM

Q14. Disadvantage of ASK compared to FSK:

A) High bandwidth B) High noise sensitivity

C) Complex receiver D) Slow speed

Q15. FSK frequency delta for f_1=1200Hz, f_0=2200Hz is:

A) 3400 Hz B) 1000 Hz

C) 500 Hz D) 2000 Hz

3. Complete Answer Key

Q#	Ans	Explanation Summary
1-5	C, B, B, A, B	AM varies amplitude; AM B=$2f_m$; Noise affects amplitude; Carson's Rule; ASK = OOK.
6-10	C, B, B, B, A	FSK amplitude fixed; μ>1 causes distortion; 2×5=10kHz; Peak amplitude fixed; ASK B ≈ R_b.
11-15	B, B, A, B, B	FSK used in modems; β=75/15=5; Fiber uses ASK; ASK noise sensitive; |2200-1200|=1000Hz.